

HARIBASKAR RAMACHANDRAN

HVAC Design Lead | MEP Design Engineer | Data Center Infrastructure Specialist

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PROFESSIONAL SUMMARY

Dynamic HVAC & MEP Design Professional with 12+ years of international experience in Data Centers, Healthcare Facilities, Metro Rail Infrastructure, Commercial Buildings, Marine Projects, and Industrial Developments. Proven expertise in HVAC system design, critical cooling infrastructure, BIM coordination, fire protection systems, tunnel ventilation, and sustainable engineering solutions. Currently working as HVAC Design Lead at Inova DC, supporting next-generation AI-ready and high-density data center environments with advanced cooling and modular infrastructure solutions. Experienced in delivering technically compliant and energy-efficient designs aligned with ASHRAE, NFPA, LEED, and international standards. Strong background in multidisciplinary coordination, project execution, BIM workflows, and global project delivery across India, UAE, Singapore, and Saudi Arabia.

CORE COMPETENCIES

HVAC System Design | Data Center Cooling Solutions | CRAC / CRAH Systems | BIM Coordination & Clash Detection | Tunnel Ventilation Systems | Healthcare HVAC Design | Fire Protection Systems | MEP System Integration | ASHRAE & NFPA Compliance | Energy Efficient Design | Revit MEP | AutoCAD | Navisworks | HAP Load Calculations | Project Coordination | Technical Documentation | BMS Integration | Mission Critical Facilities

PROFESSIONAL EXPERIENCE

HVAC Design Lead | Inova DC

Mar 2026 – Present | Bangalore, India / Global Projects

- Leading HVAC and thermal management design for AI-ready and high-density data center environments.
- Supporting modular and scalable cooling solutions including advanced airflow management and energy optimization.
- Collaborating with multidisciplinary engineering teams for mission-critical infrastructure projects across global regions.
- Driving BIM-based coordination, design reviews, and technical compliance for data center infrastructure.
- Contributing to sustainable cooling strategies aligned with next-generation AI and hyperscale data center standards.

Senior Mechanical Design Engineer | JSE Engineering Pvt Ltd

Apr 2025 – Mar 2026 | Tirunelveli, India

- Led HVAC system design for healthcare and research facilities including laboratories, operation theaters, lecture halls, and research centers.
- Ensured designs complied with ASHRAE 170, NFPA, and local healthcare authority standards.
- Reviewed contractor shop drawings, technical submittals, and equipment specifications.
- Implemented energy-efficient and sustainable HVAC solutions for healthcare environments.

Senior Mechanical Design Engineer (MEP) | BIM Global Solutions Pvt Ltd

Sep 2023 – Mar 2025 | Chennai, India

- Designed and coordinated MEPF systems for metro railway stations and tunnel ventilation infrastructure projects.
- Performed HVAC calculations including airflow, pressure drop, and ventilation analysis.
- Developed and maintained BIM models using Revit and Navisworks for multidisciplinary coordination.
- Ensured compliance with SS553, ASHRAE, IDC accessibility standards, and fire safety regulations.

Mechanical Engineer (MEP) | RIC Projects Pvt Ltd

Feb 2021 – Sep 2023 | Chennai, India

- Designed Tier-2 data center cooling systems including CRAC/CRAH units and airflow optimization solutions.
- Developed BIM LOD 400 models for HVAC, plumbing, electrical, and fire protection systems.
- Handled project coordination, prototype development, design conflict resolution, and as-built documentation.
- Supported implementation of fire suppression systems including FM200 and inert gas systems for mission-critical facilities.

Mechanical Engineer (MEP) | Sree Kartika Fabs Pvt Ltd

Jun 2017 – Oct 2020 | Vizag, India

- Designed shipboard HVAC, plumbing, propulsion support, and ventilation systems for naval projects.
- Prepared engineering calculations, technical specifications, and detailed AutoCAD drawings.
- Coordinated with marine engineers, naval architects, and multidisciplinary project teams.

Project Engineer (MEP) | Marwan Hilal Electro Mechanical LLC

Nov 2014 – May 2017 | Dubai, UAE

- Managed HVAC and MEP coordination for educational and commercial building projects.
- Integrated chilled water BTU metering systems with BMS solutions.
- Coordinated with consultants, contractors, and clients to ensure successful project execution.

Trainee Engineer | Dencil Pumps & Systems Pvt Ltd

May 2014 – Oct 2014 | Mumbai, India

- Supported senior engineers in pump system design, manufacturing processes, and technical documentation.
- Prepared basic 2D drawings and 3D models using AutoCAD and engineering software tools.
- Performed engineering calculations and assisted in maintaining BOMs and project reports.
- Contributed to product development and project execution activities for industrial systems.

MAJOR PROJECTS

- Wynn Al Marjan Island, UAE – BIM coordination, MEP integration, clash detection, and project collaboration.
- Metro Railway Stations, Singapore – Tunnel ventilation systems, MEP coordination, and BIM development.
- Red Sea International Airport, Saudi Arabia – BIM modeling and multidisciplinary MEP coordination.
- Tier-2 Data Center Projects, IIT Madras – Critical HVAC cooling systems and airflow management.
- SWISS International School, Dubai – HVAC system integration and BTU meter implementation.
- Al Mas Tower & Discovery Gardens, Dubai – Chilled water BTU integration and BMS coordination.
- Naval Dockyard Projects, Vizag – Shipbuilding HVAC and marine infrastructure support systems.

EDUCATION

Bachelor of Engineering – Mechanical Engineering

Karpaga Vinayaga College of Engineering and Technology, Chennai

Diploma in Mechanical Engineering

Lakshmi Ammal Polytechnic College

TECHNICAL SKILLS

Software: Revit MEP, AutoCAD, Navisworks, HAP, MS Project, Autodesk Design Review

Standards & Codes: ASHRAE, NFPA, LEED, SS553

Specializations: HVAC Load Calculations, Data Center Cooling, BIM 4D/5D, Fire Protection Systems, Tunnel Ventilation

ADDITIONAL HIGHLIGHTS

- 12+ years of expertise in mission-critical HVAC and MEP infrastructure projects.
- Strong exposure to international projects across UAE, Saudi Arabia, Singapore, and India.
- Hands-on experience in BIM coordination, clash detection, and multidisciplinary engineering workflows.
- Skilled in client coordination, technical leadership, and project delivery within fast-paced environments.